

Course code : **CSE3009**
Course title : **No SQL Data Bases**
Module : **2**
Topic : **3**

Columnar Data Model

Objectives

This session will give the knowledge about

- Columnar Data Model

Columnar Data Model

In column-oriented NoSQL database, data is stored in cells grouped in columns of data rather than as rows of data.

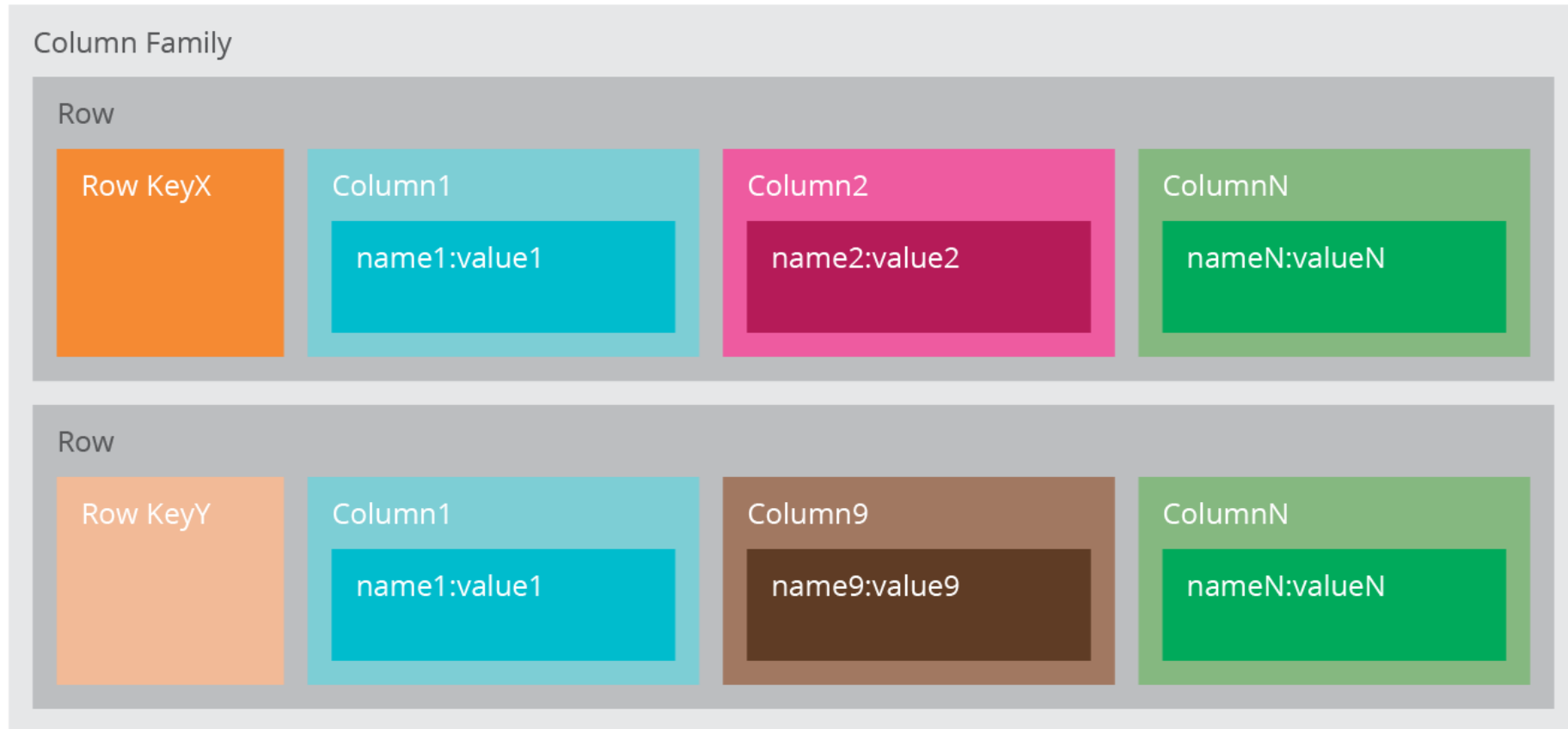
Columns are logically grouped into column families.

Column families can contain a virtually unlimited number of columns that can be created at runtime or the definition of the schema.

Read and write is done using columns rather than rows.

It can scale to manage large volumes of data

Columnar Data Model



Columnar Data Model

Column-family databases store data in **column families as rows that have many columns associated with a row key** as mentioned in previous Fig.

Each column family can be compared to a container of rows in an RDBMS table where the **key identifies the row and the row consists of multiple columns**.

The difference is that **various rows do not have to have the same columns**, and **columns can be added to any row at any time without having to add it to other rows**.

Columnar Data Model

When a column consists of a map of columns, then we have a **super column**.

A **super column** (container of columns) **consists of a name and a value which is a map of columns**.

Cassandra is one of the popular column-family databases.

Cassandra can be described as fast and easily scalable with write operations spread across the cluster.

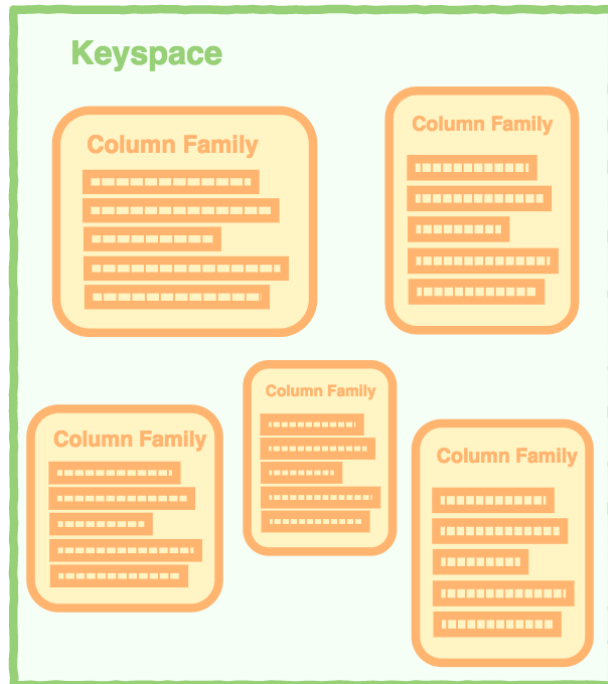
The cluster does not have a master node, so any read and write can be handled by any node in the cluster.

Columnar Data Model

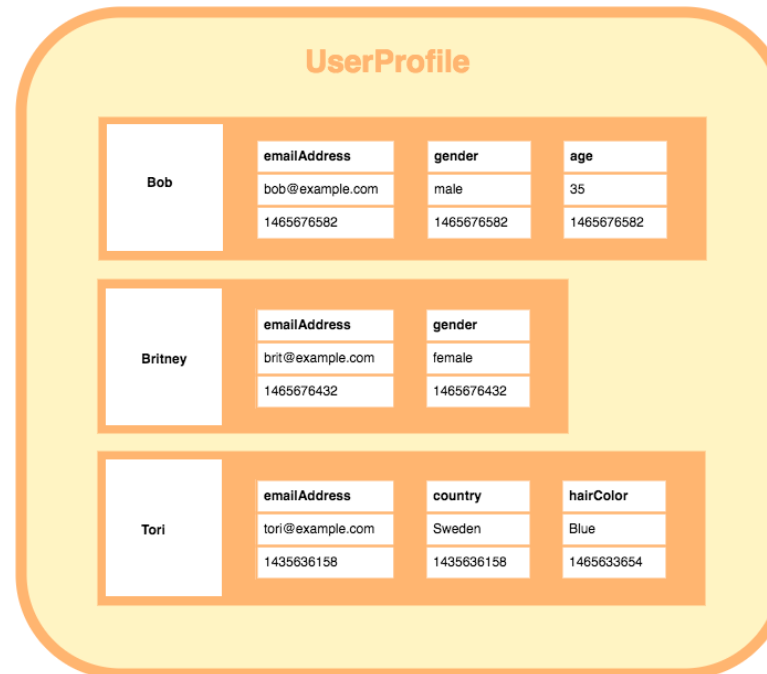
- **ColumnFamily:** ColumnFamily is a single structure that can group Columns and SuperColumns with ease.
- **RowKey:** the permanent name of the record. Keys have different numbers of columns, so the database can scale in an irregular way.
- **Keyspace:** This defines the outermost level of an organization, typically the name of the application. For example, 'InfosysDataBase' (database name).
- **Column:** It has an ordered list of elements as tuple with a name and a value defined.

Columnar Data Model

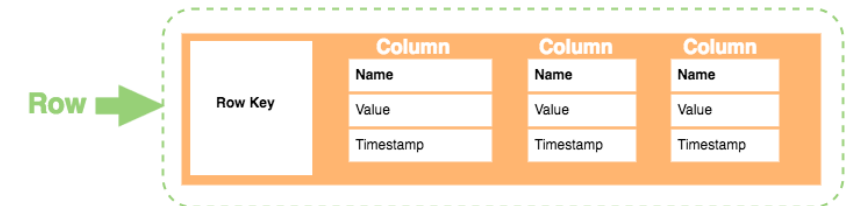
A keyspace containing column families.



A column family containing 3 rows. Each row contains its own set of columns.



how each row is constructed



Columnar Data Model

Here's a breakdown of each element in the row:

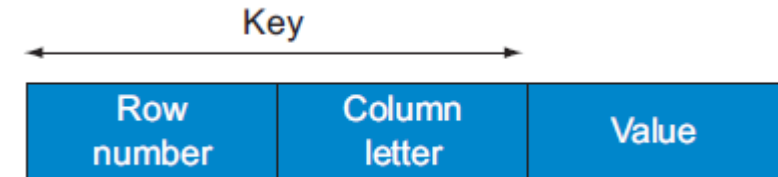
- **Row Key:** Each row has a unique key, which is a unique identifier for that row.
- **Column:** Each column contains a name, a value, and timestamp.
- **Name:** This is the name of the name/value pair.
- **Value:** This is the value of the name/value pair.
- **Timestamp:** This provides the date and time that the data was inserted. This can be used to determine the most recent version of data.

Column family basics

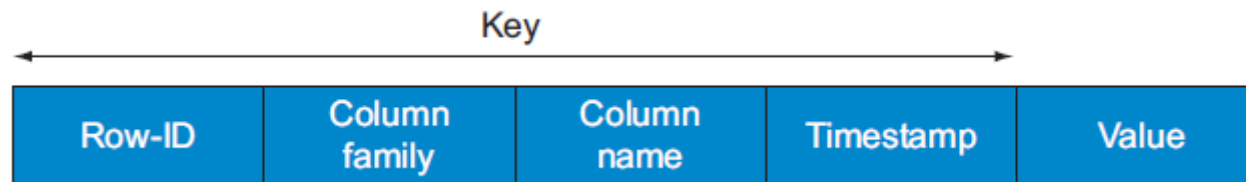
Using a row and column to address a cell.

	A	B	C
1			
2			
3		Hello World!	
4			
5			
6			

Spreadsheets use a row column pair as a key to look up the value of a cell.



In Column family stores



Columnar Data Model - Example

The best known examples are **Google's BigTable** and HBase & Cassandra that were inspired from BigTable.

For example, a **Geographic Information System (GIS)** like Google Earth might have a row ID for the longitude portion of a map and use the column name for the latitude of the map.

If you have one map for each square mile on Earth, you could have 15,000 distinct row IDs and 15,000 distinct column IDs.

Columnar Data Model - Example

BigTable, is a high performance, compressed and proprietary data storage system owned by Google. It has the following attributes:

- Sparse – some cells can be empty
- Distributed – data is partitioned across many hosts
- Persistent – stored to disk
- Multidimensional – more than 1 dimension
- Map – key and value
- Sorted – maps are generally not sorted but this one is

Columnar Data Model - Example

A 2-dimensional table comprising of rows and columns is part of the relational database system.

City	Pincode	Strength	Project
Noida	201301	250	20
Cluj	400606	200	15
Timisoara	300011	150	10
Fairfax	VA 22033	100	5

Columnar Data Model - Example

For previous RDBMS table a BigTable map can be visualized as shown below.

```
{
  InfosysNoida: {
    address: {
      city: Noida
      pincode: 201301
    },
    details: {
      strength: 250
      projects: 20
    }
  }
}
```

```
{
  InfosysCluj: {
    address: {
      city: Cluj
      pincode: 400606
    },
    details: {
      strength: 200
      projects: 15
    }
  },
}
```

```
{
  InfosysTimisoara: {
    address: {
      city: Timisoara
      pincode: 300011
    },
    details: {
      strength: 150
      projects: 10
    }
  }
}
```

```
{
  InfosysFairfax : {
    address: {
      city: Fairfax
      pincode: VA
      22033
    },
    details: {
      strength: 100
      projects: 5
    }
  }
}
```

Columnar Data Model - Example

- The outermost keys InfosysNoida, InfosysCluj, InfosysTimisoara and InfosysFairfax are analogues to rows.
- ‘address’ and ‘details’ are called column families.
- The column-family ‘address’ has columns ‘city’ and ‘pincode’.
- The column-family details’ has columns ‘strength’ and ‘projects’.
- Columns can be referenced using CloumnFamily.

Google’s BigTable, HBase and Cassandra are the most popular column store based databases.

Columnar Data Model – Example in Cassandra

Insert into InfosysNoida.address(city,pincode) values('Noida',201301);

Insert into InfosysNoida.details(strength,projects) values(250,20);

Update InfosysNoida.address Set city='UP' Where pincode=201301;

Delete from InfosysNoida.address where pincode=201301;

Benefits of Column Store Databases

The column family approach of **using a row ID and column name as a lookup key** is a flexible way to

- store data,
- gives you benefits of higher scalability and availability, and
- saves you time and hassles

when adding new data to your system.

Column family systems have **automatic failover built in to detect failing nodes and algorithms to identify corrupt data.**

The **advanced hashing and indexing tools** such as Bloom filters are used in column data stores **to perform probabilistic analysis on large data sets.**

Benefits of Column Store Databases

Higher scalability:

Column family systems are scalable by nature. You can add more data to your system, you can add more cluster to the computing cluster.

Higher availability:

The **server that holds part of the sparse matrix crashes**, other computers are available to provide the data service for those cells. Because

- column family systems use efficient communication, the cost of replication is lower
- the lack of join operations allows you to store any portion of a column family matrix on remote computers

Benefits of Column Store Databases

Easy to add new data

- Like the key-value and graph stores, a key feature of the column family store is that **you don't need to fully design your data model before you begin inserting data.**
- But there are a couple constraints:
 - **Your groupings of column families should be known in advance,** but row IDs and column names can be created at any time.
 - Column family systems also don't support standard SQL queries for real-time data access. **They may have higher-level custom API written in a procedural language.**

Benefits of Column Store Databases

Some other key benefits of columnar databases include:

- **Compression:** Column stores are very efficient at data compression and/or partitioning.
- **Aggregation queries:** Due to their structure, columnar databases perform particularly well with aggregation queries (such as SUM, COUNT, AVG, etc).
- **Fast to load and query:** Columnar stores can be loaded extremely fast. A billion row table could be loaded within a few seconds. You can start querying and analyzing almost immediately.

Summary

This session will give the knowledge about

- Columnar Data Model